

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1013

Boyles 100 kV
October 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1013. The Interconnection Customer proposes a 81.6 MW solar photovoltaic facility interconnecting at the Boyles 100 kV substation in the Ohio area. The request was received on June 06, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 371 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,141,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The CANAL ROAD-WEEDPATCH SUBSTATION 230kV is overloaded to 113.2% of its 1,187 MVA emergency rating for the loss of the TREGO-BEAVR 500kV. The project contributes 7.3% DFAX (6.0 MW) to this overload.

The NORWICH 230-1 kV XFMR Bank 1 is overloaded to 110.1% of its 995 MVA emergency rating for the loss of the BARRETT-NORWICH 500kV. The project contributes 6.3% DFAX (5.1 MW) to this overload.

For the loss of TREGO-BEAVR 500kV L, the GALION GENERATING STA-MYRTLE POINT 230kV loads to 109.9% of its 1,298 MVA rating. Project DFAX: 7.9%.

For the loss of RAVENNA-HYLEBOS 500kV, the CHAMBERLIN SUBSTATION-IMPERIAL 230kV line loads to 109.0% of its 1,299 MVA rating. Project DFAX: 11.4%.

For the loss of TROY ENERGY LLC-TREGO 500kV, the TREGO-SWAN VALLEY 500kV loads to 106.7% of its 4,453 MVA rating. Project DFAX: 7.5%.

For the loss of TREGO-TROY ENERGY LLC 500kV L, the TREGO 230/1 kV Auto loads to 106.2% of its 829 MVA rating. Project DFAX: 20.8%.

For the loss of GALION GENERATING STATION-TREGO 500kV, the MYRTLE PT-GEORGE NEAL SOUTH SUBSTATION 230kV L loads to 104.2% of its 974 MVA rating. Project DFAX: 5.6%.

ZONA LIBRE 6007-GEORGE NEAL SOUTH 230kV: 103.9% loading (1,001 MVA) following the outage of the TREGO-GALION GENERATING STA 500kV; DFAX = 5.6%.

For the loss of TREGO-GALION GENERATING STA 500kV, the ZONA LIBRE 6007-KAMPSVILLE 230kV loads to 103.7% of its 1,028 MVA rating. Project DFAX: 5.6%.

For the loss of TREGO-GALION GENERATING STA 500kV Line, the MORENO TAP-KAMPSVILLE 230kV Line loads to 103.6% of its 1,044 MVA rating. Project DFAX: 5.6%.

For the loss of TREGO-BEAVR 500kV, the WALDO 500-230 kV Auto loads to 103.1% of its 1,248 MVA rating. Project DFAX: 10.4%.

For the loss of TREGO-BEAVR 500kV, the NORWICH-BARRETT 500kV line loads to 103.1% of its 3,374 MVA rating. Project DFAX: 11.9%.

For the loss of TROY ENERGY LLC-TREGO 500kV, the HIGHGATE-TREGO 230kV loads to 101.2% of its 652 MVA rating. Project DFAX: 19.1%.

For the loss of GRAY'S FLAT-E METHUEN 230kV line, the MADELINE-EAST METHUEN 230kV L loads to 100.8% of its 1,495 MVA rating. Project DFAX: 8.0%.

For the loss of BEAVER-TREGO 500kV Line, the BARRETT-WOLFE 500kV loads to 100.8% of its 3,201 MVA rating. Project DFAX: 8.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CANAL ROAD-WEEDPATCH SUBSTATION 230kV	Rebuild line with higher-capacity conductor	\$11,196,000	2% (\$226,000)
NORWICH 230-1 kV XFMR #1	Replace transformer with higher-rated unit	\$6,108,000	2% (\$155,000)
MYRTLE PT-GALION GENERATING STA 230kV Line	Reconductor line	\$7,343,000	2% (\$174,000)
CHAMBERLIN SUBSTATION-IMPERIAL 230kV line	Reconductor line	\$4,686,000	2% (\$114,000)
SWAN VALLEY-TREGO 500kV	Reconductor line	\$95,411,000	1% (\$993,000)
TREGO 230/1 kV Auto	Replace transformer with higher-rated unit	\$12,836,000	5% (\$660,000)
MYRTLE PT-GEORGE NEAL SOUTH 230kV L	Reconductor line	\$4,131,000	2% (\$91,000)
ZONA LIBRE 6007-GEORGE NEAL S 230kV	Reconductor line	\$12,262,000	2% (\$270,000)
ZONA LIBRE 6007-KAMPSVILLE 230kV line	Reconductor line	\$9,500,000	2% (\$209,000)
MOREO TAP-KAMPSVILLE 230kV	Reconductor line	\$12,742,000	2% (\$281,000)
WALDO 500-230 kV Transformer Bank	Replace transformer with higher-rated unit	\$37,897,000	4% (\$1,396,000)
NORWICH-BARRETT 500kV line	Reconductor line	\$29,798,000	2% (\$666,000)
HIGHGATE-TREGO 230kV	Reconductor line	\$23,165,000	8% (\$1,957,000)
EAST METHUEN-MADELINE SUBSTATION 230kV line	Reconductor line	\$9,954,000	3% (\$273,000)
WOLFE-BARETT 500kV L	Reconductor line	\$62,768,000	3% (\$1,676,000)
	Total allocated to project		\$9,141,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.